

Article

Integrating Hydraulic Properties into Irrigation Management of Industrial Hemp (*Cannabis sativa* L., ‘Felina 32’) Under Mediterranean Conditions

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Abstract

Industrial hemp (*Cannabis sativa* L.) is versatile and rapidly developing, offering new prospects to producers as a multipurpose crop, yet limited water availability in the Mediterranean area due to climate change makes its sustainable management challenging. Although the plant’s water requirements have been studied, a significant gap remains regarding irrigation management based on the hydraulic properties that govern water movement. The present study elucidates the role of soil hydraulic parameters in water dynamics within the rhizosphere of industrial hemp (*Cannabis sativa* L., ‘Felina 32’). For this purpose, a pot experiment of three irrigation treatments (100% FC, 80% FC, 60% FC; FC is the field capacity) was set up using two different soil types (clay loam CL and sandy clay loam SCL). SCL soil showed a higher C_{max} of about 4 cm^{-1} compared to the C_{max} of 0.11 cm^{-1} of CL soil, but dropped drastically within a narrow frame of soil moisture. CL soil resulted in about 12-fold higher diffusivity ($D_{max} \approx 0.23\text{ cm}^2\text{ min}^{-1}$) within a wider range of soil moisture compared to the SCL soil ($D_{max} \approx 0.02\text{ cm}^2\text{ min}^{-1}$), which facilitated water redistribution at CL, allowing the plant to maximize its water uptake, even at the lowest water input. As a result, the CL soil allowed more flexible scheduling and in contrast, SCL soil necessitated a high frequency irrigation protocol. The integration of hydraulic properties into irrigation planning revealed the potential of CL to apply water to plants efficiently across full and deficit irrigation, showing the peak performance of the irrigation water use efficiency (IWUE) (0.929 g/mm) under the 60% FC regime. The findings provide a framework for linking soil physics–agricultural hydraulics with irrigation strategies in controlled environments.

Keywords: agricultural hydraulics; soil physics; soil–water interaction; precise irrigation; hydraulic capacity; hydraulic diffusivity; soil water characteristic curve (SWCC); irrigation water use efficiency (IWUE)



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1. Introduction

Over the last few years, Mediterranean and semi-arid regions have often suffered from water scarcity due to climate change. The impact of climate change on crop selection is big, as farmers are increasingly turning to more drought-tolerant crops. Lately, more farmers have given their attention to industrial hemp (*Cannabis sativa* L.), a crop that is widely known for its pharmaceutical characteristics, along with cloth production ability and

biofuel [1]. One of the main advantages of hemp cultivation is the moderate to low input requirements, making it an important crop for sustainable agriculture [2–5]. Moreover, the Felina 32 variety has a low tetrahydrocannabinidiol (THC) level and is usually used for fiber [6,7]. Thus, Felina 32 has attracted many researchers to investigate its response to various hydraulic conditions, along with the air and soil temperature, soil contamination and climate conditions for either fabric or bioenergy use [8–10].

Water is the most vital water resource; however, its scarcity due to the climate crisis restricts crop productivity. In recent years, hazards have been increasing, as rainfall more often leads to floods, while on the other hand, droughts are expanding. In areas where water is scarce, such as the Mediterranean, sustainable water resource management combined with crop production is a critical challenge and a key solution for the future [11,12]. Crop productivity is connected with the texture and aggregation of the soil [13], while the soil's hydraulic properties play a decisive role in irrigation planning and practices [14–19]. Hydraulic properties such as hydraulic conductivity, diffusivity, capacity, infiltration rate, etc., are linked to irrigation planning and scheduling and play a vital role in influencing plant behavior under different irrigation regimes. In addition, hydraulic properties are crucial for the horizontal and vertical movement of the irrigation water, with a great impact on the irrigation dose and duration [20]. Among the crucial hydraulic properties, field capacity (FC) is the ideal value for soil moisture for crops [11] and due to its importance, the present research uses FC as the upper level of soil moisture, while irrigation treatments were designed as FC ratios. FC varies according to the soil texture, because the clay, silt and sand percentages of the soil directly affect the water retention and, therefore, the water availability, which in turn regulates the physiological characteristics, such as height and leaf development, which are linked to biomass accumulation [21].

Lately, several researchers focused on water-consumption-related factors, such as crop coefficients (K_c) and crop evapotranspiration (ET_c), along with the agronomic response of the crop to the water deficit and temperature [21–23], but there is still a significant lack of investigation into the effect of the hydrodynamics of the soil–water dipole, along with the hydraulic properties, on irrigation decisions and hemp's behavior. The response of industrial hemp to controlled hydro-stress remains under investigation, particularly when the texture, structure and hydraulic properties are factors under consideration. Moreover, as deficit irrigation enhances the irrigation water use efficiency (IWUE) with simultaneous water resource conservation [2,4,22], controlled water stress may play a key role in the cultivation of industrial hemp in areas where droughts have become increasingly frequent and water scarcity is the harsh reality. To that direction, considering the hydraulic properties along with the soil's physicochemical characteristics, we can achieve precise and efficient water application with a simultaneous undamaged yield.

The objective and innovation of the present study is to elucidate the role of hydraulic properties in governing water dynamics in the root zone of industrial hemp. By integrating high-resolution hydraulic properties such as the hydraulic capacity (or specific water capacity, specific moisture capacity, or differential water capacity) $C(\theta)$; hydraulic diffusivity $D(\theta)$; hydraulic conductivity $K(\theta)$, including van Genuchten's model parameters; SWCC and infiltration into irrigation decisions, this research highlights how soil physics and agricultural hydraulics are connected to irrigation strategies. The study provides a mechanistic framework for understanding the role of soil water redistribution and hydraulic behavior in water management. Thus, this research intends to contribute to the current understanding of irrigation water management by presenting an additional perspective that promotes crop protection and the efficient use of water resources. A containerized cultivation system was intentionally selected to ensure a controlled hydraulic environment, which allowed for the monitoring of the water balance, avoiding deep percolation and runoff, minimizing the

spatial heterogeneity of soil, and avoiding external factors such as competition from weeds and soil obstacles, while the isolation of soil was achieved by comparing two different soil types under the same environmental conditions.

2. Materials and Methods

2.1. Experimental Design

The experiment was implemented in Velestino, Greece, at the farm of the University of Thessaly (39°23'44" N, 22°45'26" E, alt. 70 m) during the Spring–Summer season of 2022. Two soil types were used: one taken directly from the farm (soil A) and the other (soil B) was a mixture of sand and farm soil in a 1:10 ratio. Thirty pots were filled with both soil types (half filled with soil A and the remaining half filled with soil B). Industrial hemp (*Cannabis Sativa* L., 'Felina 32') seeds were planted in 20 L volume pots, each of which was filled with 10 L of soil. To prevent seeds from microbiological impact, before seeding, NaClO was used for ten minutes over the seeds [6]. Sowing followed the rule of 10 seeds per pot and the plants were later thinned to three per pot. The experimental design consisted of 2 soil types (A, B) $\times$ 3 irrigation treatments (I1, I2, I3) $\times$ 5 replications = 30 pots. During the experimental period, one plant with the B-I1 treatment showed symptoms of disease, leading to its exclusion. To ensure a strictly balanced experimental design, we standardized the dataset to 4 replicates per treatment. Irrigation water was applied, consisting of 3 irrigation treatments, as follows: I1: 100% FC (0% residual) defined as the control, I2: 80% FC (20% residual), and I3: 60% FC (40% residual). Water was applied using the spaghetti tube irrigation system and the flow rate was 10 mm/h. The soil moisture was monitored daily by Delta-T SM 150. The plant's quantitative characteristics (length (L), biomass (m) and dry biomass (dry m)) were measured, using a measuring scale and a digital weighing scale. At each pot, the average of the three plants was recorded (for the characteristics under research (L, m and dry m)), and the average of the replications of each treatment was also recorded. Figure 1 presents the experimental procedure during two representative stages of plant growth.



Figure 1. The experimental procedure at two representative stages of the experiment (a) early stage; (b) installation of the irrigation system.

For each soil, the critical soil moisture (θ_c) was determined and used as the lowest threshold of soil moisture, following Equation (1) [24,25]:

$$\theta_c = \theta_{FC} - p(\theta_{FC} - \theta_{PWP}) \quad (1)$$

where θ_c is the critical soil moisture, θ_{FC} is the field capacity of the soil, θ_{PWP} is the permanent wilting point of the soil and p is the maximum allowable depletion (MAD), equal to 0.6, which was selected according to the hydraulic properties obtained.

2.2. Soil Physicochemical Properties

The soil samples were taken to the laboratory and dried in the oven at 105 °C, and the soil characteristics have been analyzed [13,26–28]. The samples were sieved using a 2 mm pore sieve and underwent chemical analyses to estimate the pH, the electrical conductivity (EC), the organic matter (OM), the CaCO_3 and the soil particle distribution [6,28,29]. The soil pH and electrical conductivity (EC) were assessed in a soil:water (1:1) mixture; the organic matter content was measured using the Walkley Black method, using K_2CrO_4 as an oxidizing factor. The content of CaCO_3 was measured by a proper calcimeter and the particle size distribution of the soil was estimated by the Bouyoucos method. In the experimental procedure of the current study, the same original soil (A) and soil ratio (B) with [13] were used.

2.3. Hydraulic Properties

The soil samples were delivered to the laboratory to measure the following hydraulic properties:

- Field capacity (FC).
- Permanent wilting point (PWP).
- Soil moisture at saturation (θ_s).
- Bulk density (ρ_b).
- Cumulative infiltration $[I(t)]$ and infiltration rate $[i(t)]$.
- Soil water characteristic curves $[SWCC\ h(\theta)]$.

The van Genuchten equation [30] was used to simulate the SWCC:

$$\Theta = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \frac{1}{[1 + (\alpha h)^n]^m} \quad (2)$$

where θ_s is soil water content at saturation and θ_r is the residual water content, while a , m , and n are parameters which have to do with the shape and the curvature of the SWCC. The Kostiaikov [31] equation was used to determine the cumulative infiltration functions for both soils being researched.

For the determination of FC, PWP, and SWCC, the experimental device used at the laboratory was a pressure plate apparatus (Soil Moisture Equipment Corp., Goleta, CA USA), while a digital weighing scale was used to calculate θ_s . The soil moisture thresholds (FC and PWP) were determined by applying 0.33 bar and 15 bar tensions to the saturated soil samples, correspondingly. For the determination of the SWCC, the tension range was 0.05 to 15 bars for soil A (CL) and 0.04 bar to 15 bars for soil B (SCL). Infiltration was measured by using transparent columns of 50 cm height and 6 cm inner diameter, measuring the incoming water at certain time intervals. RETC v. 6.02 software was used to simulate the SWCC and the hydraulic properties of hydraulic conductivity $K(\theta)$ and hydraulic capacity $C(\theta)$, which are equal to a derivative of soil moisture in terms of suction $d\theta/dh$ and hydraulic diffusivity $D(\theta)$, which are equal to $K(\theta)/C(\theta)$.

2.4. Meteorological Data

Meteorological data were obtained from the meteorological station of the University of Thessaly experimental farm located in close proximity to the study site.

2.5. Statistical Analysis

SPSS (IBM SPSS Statistics v29.0. Armonk, NY, USA: IBM Corp) was used for statistical analysis. General linear models were used for the analysis of variance (ANOVA) and Tukey post hoc tests were used for pairwise comparisons. The mean difference is significant at $\alpha = 0.05$ level ($p < 0.05$).

2.6. Python Code

Python 3.10 code was used for certain data analysis. ChatGPT5.0 (OpenAI, San Francisco, CA, USA) was used as a coding assistant to help structure portions of Python 3.10 code. Python code was utilized exclusively for advanced data visualization and density analysis. All code, data handling, and figures were verified, validated, and edited by the authors, using original experimental data.

3. Results

The Mediterranean climate is well known for its very warm summers, along with mild winters. The average values of the air temperature during the growth period varied between 15.2 °C and 27.9 °C, while the values of precipitation during the same period varied between 1 mm and 57.6 mm (Figure 2).

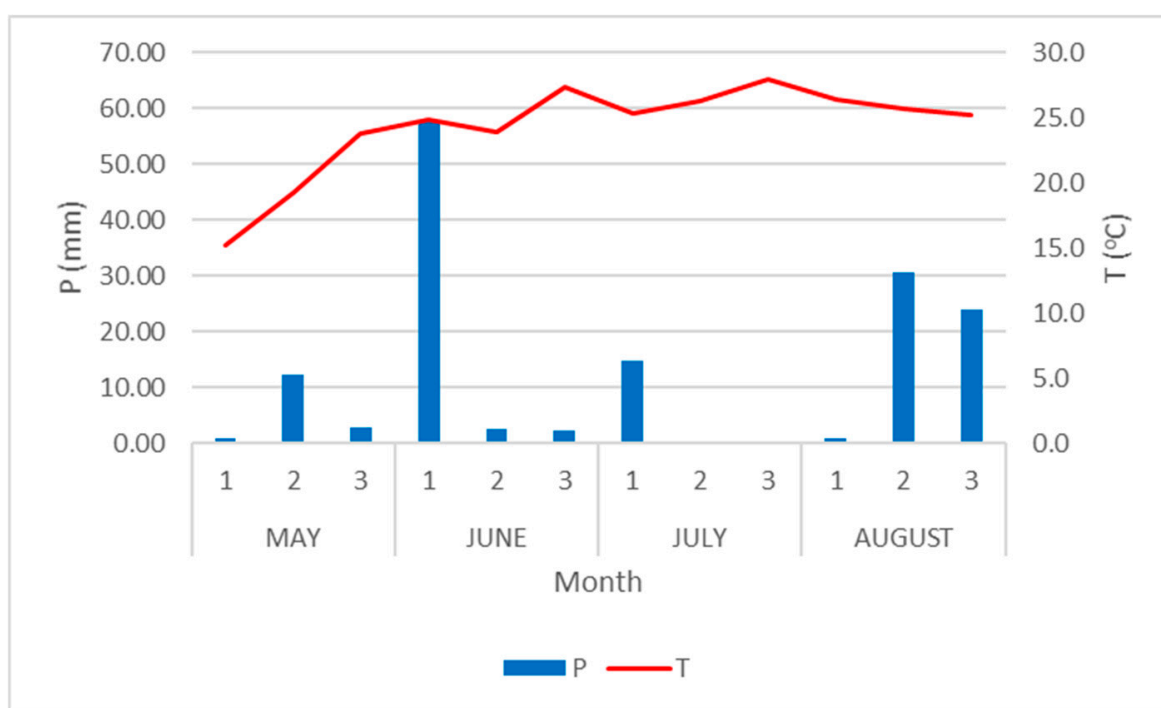


Figure 2. Meteorological data [precipitation P (mm) and temperature T (°C)].

Table 1 presents the physicochemical characteristics of the soil samples under investigation. Soil A was characterized as clay loam (CL) and soil B was characterized as sandy clay loam (SCL) [6,13]. The original farm soil (soil A) was classified as calcixerollic xerochrept [7], while the classification was the same for the mixed soil, B. Both soils were alkaline. The values of the critical soil moisture for each soil sample were found using Equation (1): $\theta_c^{CL} = 0.20 \text{ cm}^3 \text{ cm}^{-3}$ ($= 20.0\% \text{ v/v}$) and $\theta_c^{SCL} = 0.164 \text{ cm}^3 \text{ cm}^{-3}$ ($= 16.4\% \text{ v/v}$).

Table 1. Physicochemical parameters of soil types A and B [13].

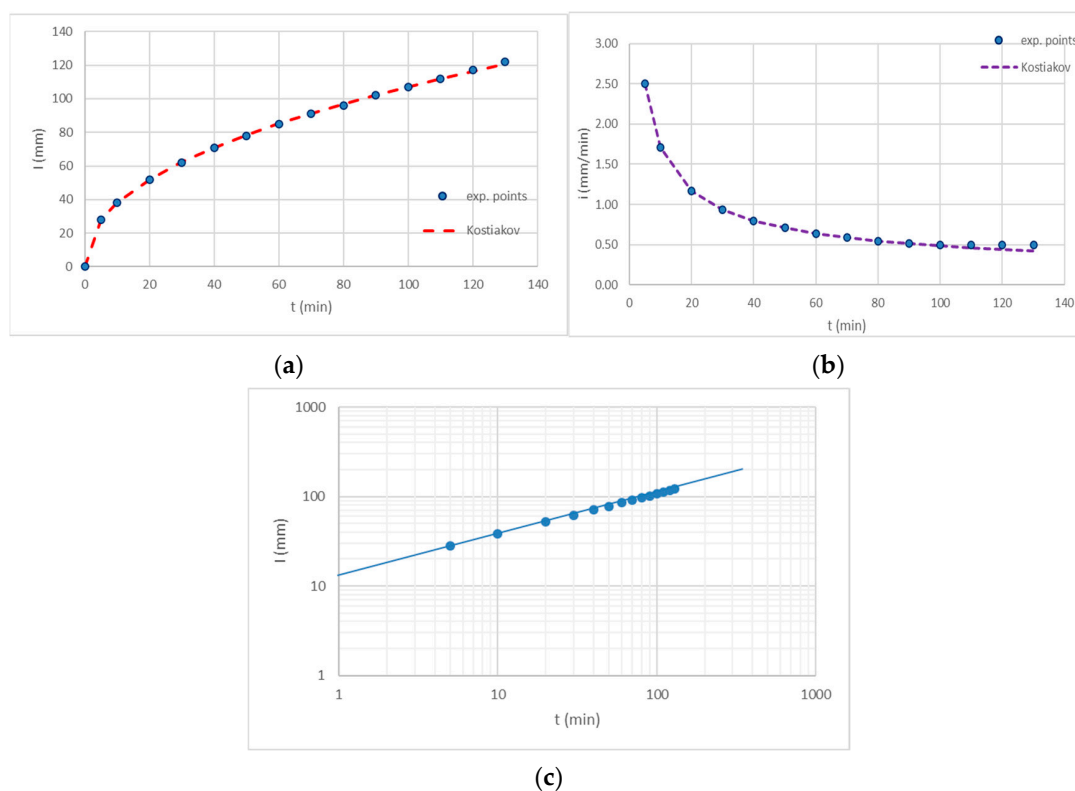
Physicochemical Characteristics	Soil A (CL)	Soil B (SCL)
pH (1:1)	7.45 ± 0.7	7.31 ± 0.5
EC (μS cm ⁻¹)	475.7 ± 6.5	357.0 ± 3.2
OM (%)	2.1 ± 0.4	1.4 ± 0.3
CaCO ₃ (%)	9.2 ± 0.9	7.8 ± 1.2
Clay (%)	32.0 ± 1.2	28.2 ± 3.2
Silt (%)	40.0 ± 2.2	21.3 ± 2.1
Sand (%)	28.0 ± 2.2	50.5 ± 2.8
ρ _b (g cm ⁻³)	1.40 ± 0.02	1.60 ± 0.02
FC (cm ³ cm ⁻³)	0.320 ± 0.005	0.260 ± 0.005
PWP (cm ³ cm ⁻³)	0.120 ± 0.005	0.100 ± 0.005
θ _s (cm ³ cm ⁻³)	0.42 ± 0.01	0.37 ± 0.01

The two soils used for the present research were characterized as clay loam (soil A) and sandy clay loam (soil B). The particle size analysis indicated that soil B is a “light” soil, while soil A is cohesive, with a “heavier” texture. Thus, soil A has a stronger ability to retain water than soil B: a behavior that was reflected in the hydraulic properties under investigation (Table 1).

Figure 3 shows (a) the cumulative infiltration curve, (b) the infiltration rate curve and (c) the logarithmic graph of $I(t)$ for soil A. The estimated equations [31], using the graph method, are presented below:

$$I = 13.5 t^{0.45} \text{ (mm, min)} \quad (3)$$

$$i = 6.075 t^{-0.55} \text{ (mm/min, min)} \quad (4)$$

**Figure 3.** Infiltration curves for soil A: (a) Cumulative infiltration versus time; (b) infiltration rate versus time; and (c) cumulative infiltration versus time (log-log).

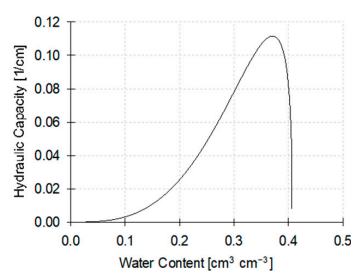
Equation (1) was used to estimate the start time of the irrigation, while the duration of the irrigation was determined by taking into consideration the difference between the field capacity and critical soil moisture, and Equations (3) and (4) were implemented as a protective buffer to prevent from overirrigation and runoff.

Table 2 shows the van Genuchten parameters, Figure 4 shows (a) the hydraulic capacity with the soil moisture curve, (b) the hydraulic conductivity versus soil moisture, (c) the diffusivity versus soil moisture, (d) the pressure head (log) with soil moisture curve and (e) the soil water characteristic curve (SWCC) approximated with van Genuchten's model, for soil A.

Table 2. van Genuchten parameters for soil A.

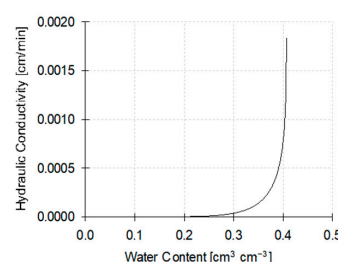
Parameter	Soil A (CL)
θ_r ($\text{cm}^3 \text{cm}^{-3}$)	0.03
θ_s ($\text{cm}^3 \text{cm}^{-3}$)	0.42
α (cm^{-1})	1.35000
n	1.53000
m	0.34600

Hydraulic Properties: C vs. Theta



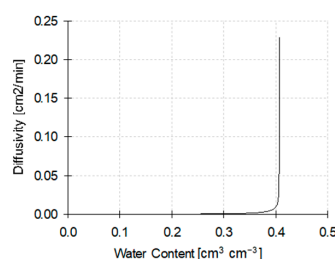
(a)

Hydraulic Properties: K vs. Theta



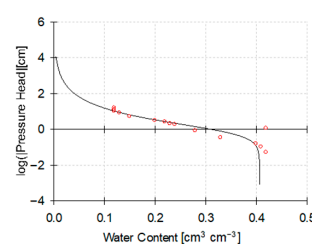
(b)

Hydraulic Properties: D vs. Theta

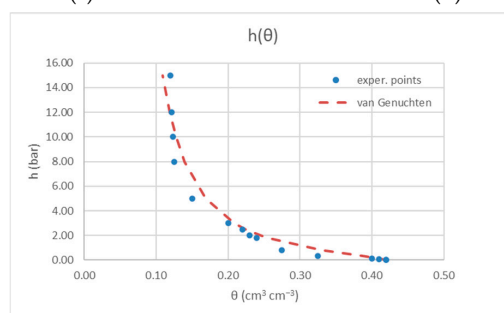


(c)

Hydraulic Properties: log h vs. Theta



(d)



(e)

Figure 4. Hydraulic properties for soil A: (a) hydraulic capacity versus soil moisture; (b) hydraulic conductivity versus soil moisture; (c) diffusivity versus soil moisture; (d) SWCC (log), where the experimental points are indicated with red dots; and (e) SWCC experimental curve and van Genuchten approximation.

Table 3 presents the van Genuchten parameters, and Figure 5 shows (a) the cumulative infiltration curve, (b) the infiltration rate curve and (c) the logarithmic graph of $I(t)$ for soil B. The estimated equations are presented below:

$$I = 22.5 t^{0.39} \text{ (mm, min)} \quad (5)$$

$$i = 8.775 t^{-0.61} \text{ (mm/min, min)} \quad (6)$$

Table 3. van Genuchten parameters for soil B.

Parameter	Soil B (SCL)
$\theta_r \text{ (cm}^3 \text{ cm}^{-3}\text{)}$	0.01
$\theta_s \text{ (cm}^3 \text{ cm}^{-3}\text{)}$	0.37684
$\alpha \text{ (cm}^{-1}\text{)}$	6.88475
n	1.26257
m	0.207965

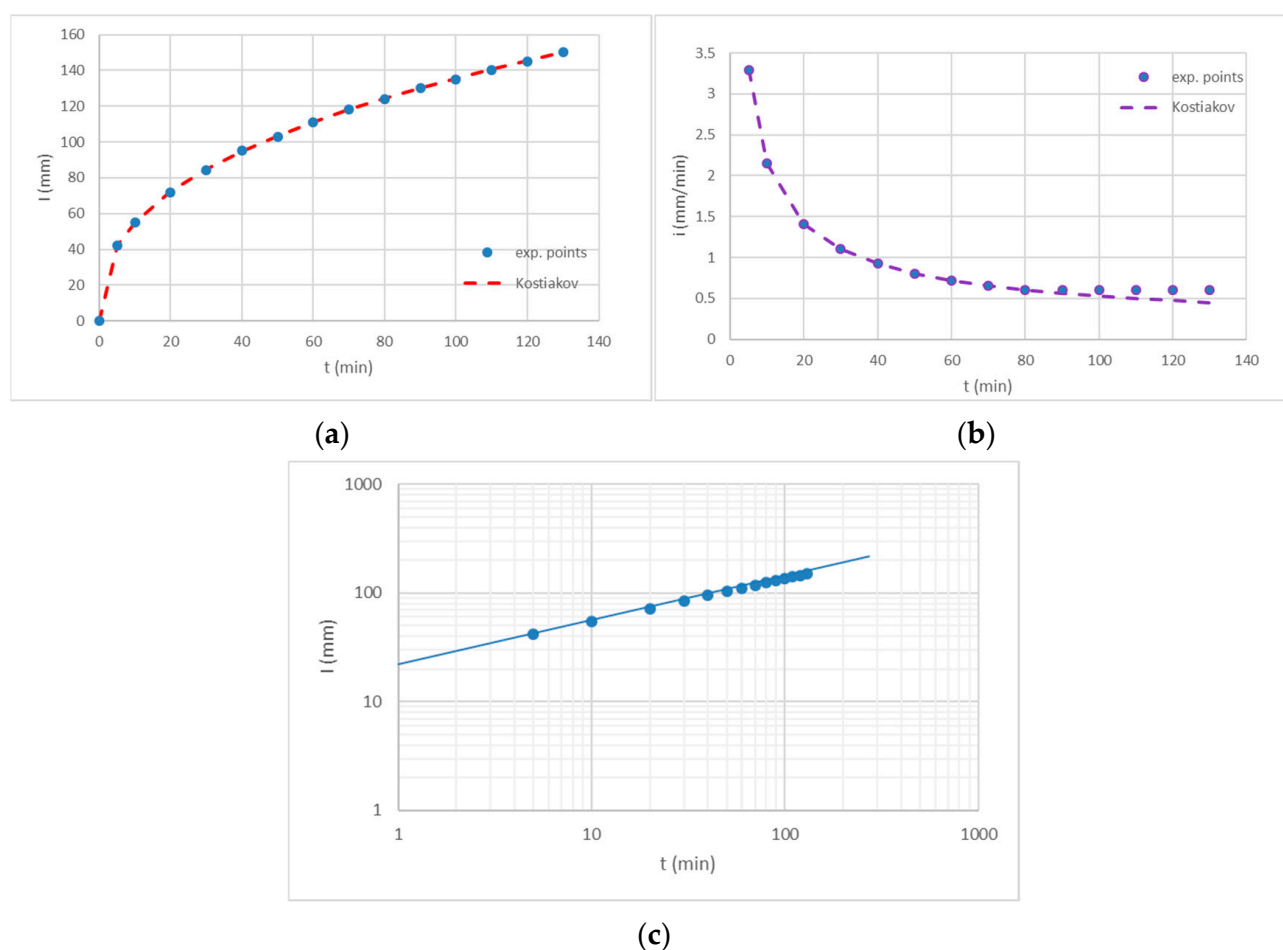


Figure 5. Infiltration curves for soil B: (a) Cumulative infiltration versus time; (b) infiltration rate versus time; and (c) cumulative infiltration versus time (log-log).

The infiltration curves of both soils were crucial for the decision of the spaghetti irrigation system and the flow rate, within the limits of final infiltrability. Soil A had initial infiltrability of (i) about 2.5 mm/min equal to 150 mm/h, while the final stable value was 0.5 mm/min or 30 mm/h. Soil B showed 3.29 mm/min or 197.4 mm/h initial infiltrability, while the final value was 0.6 mm/min or 36 mm/h. Thus, the decision to have a 10 mm/h

flow rate ensured the water absorption, avoiding ponding or losses due to overflow and runoff. The selection of the flow rate has been fixed according to the infiltration data of soil A in order to ensure hydraulic security for both soils. Figure 6 shows the estimated hydraulic parameters, along with the SWCC and van Genuchten approximation for soil B.

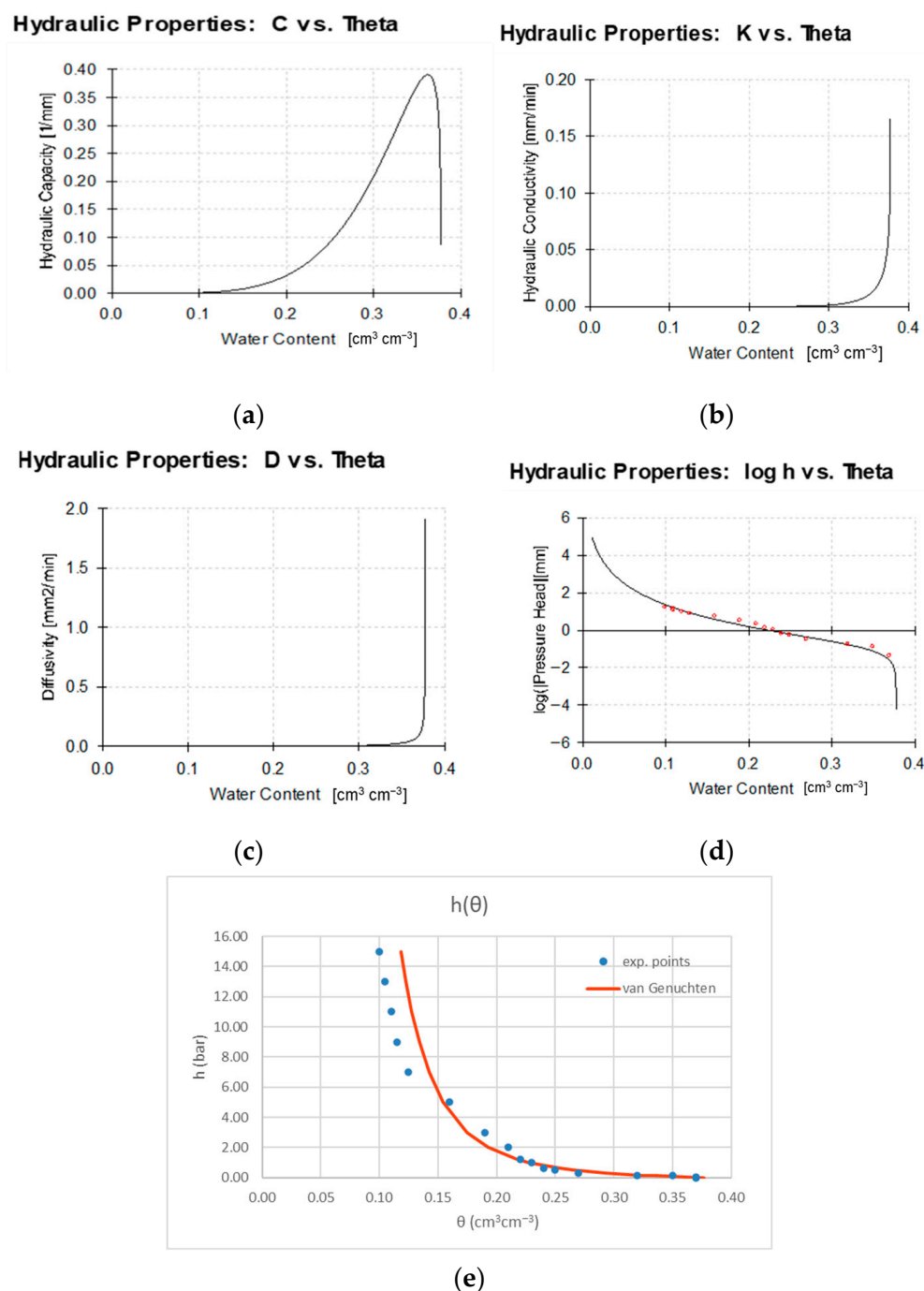


Figure 6. Hydraulic properties for soil B: (a) Hydraulic capacity versus soil moisture; (b) Hydraulic conductivity versus soil moisture; (c) Diffusivity versus soil moisture; (d) SWCC (log) where the experimental points are indicated with red dots; and (e) SWCC experimental curve and van Genuchten approximation.

In Figures 4 and 6, the differences in hydraulic properties such as hydraulic capacity, conductivity and diffusivity, along with the SWCC for both soils, are presented, where soil A (Figure 4) shows greater available soil moisture and lower maximum hydraulic capacity, but within a greater range of soil water content, than soil B, while the diffusivity values of

soil A were greater than those of soil B. The unsaturated hydraulic conductivity of soil B was significantly higher than soil A within a narrower soil moisture range, indicating the rapid drainage of the macropores in the sandy clay loam soil.

Temporal soil moisture variations are presented for all treatments and for both soils in Figure 7, where the thresholds of FC, PWP and θ_c are also shown. Figure 8 presents irrigation along with precipitation and temperature for all treatments for both soils. The divergent irrigation schedules indicate the different hydraulic behavior of each soil type, ensuring that water application was optimized according to their storage and transmission characteristics.

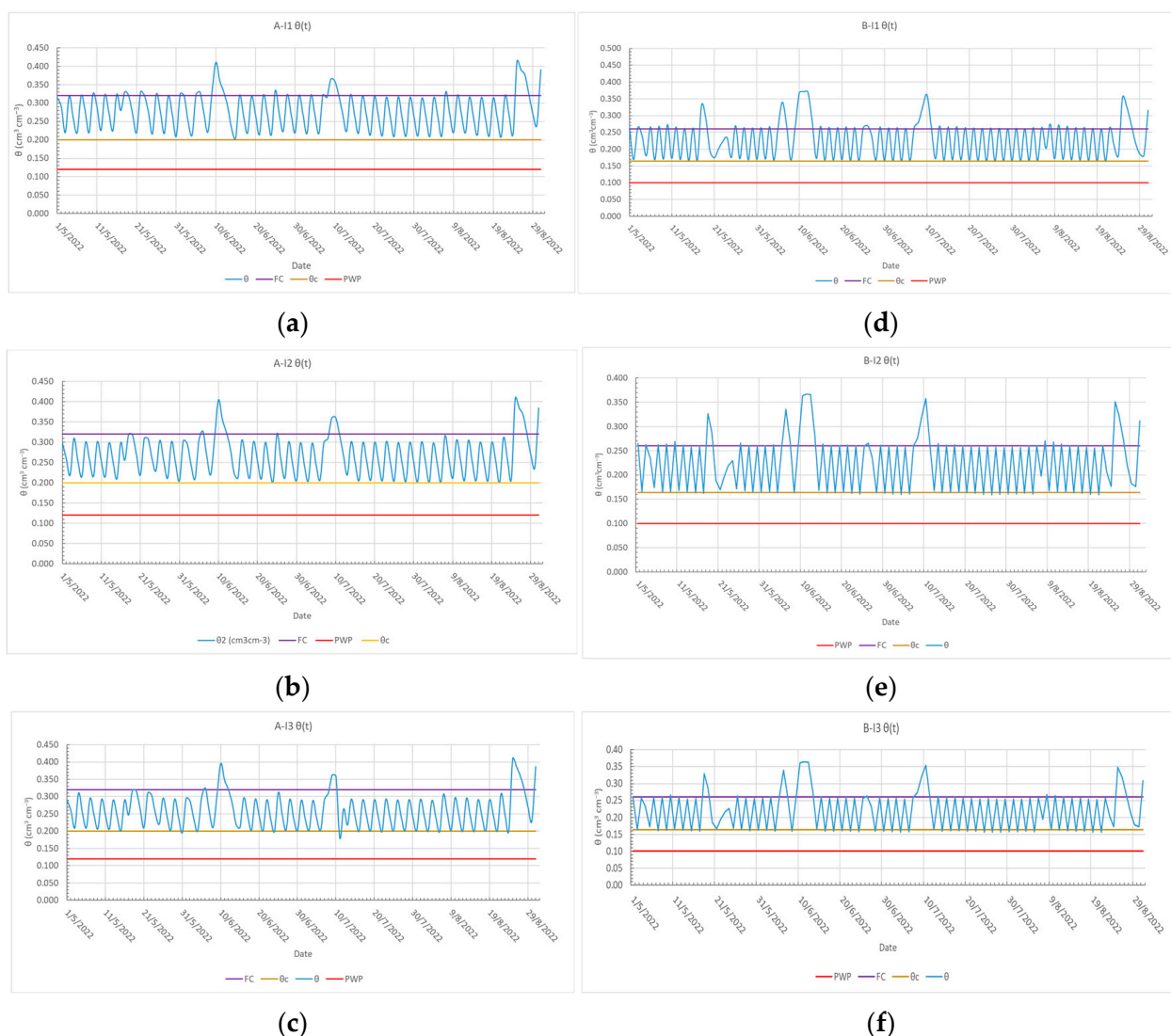
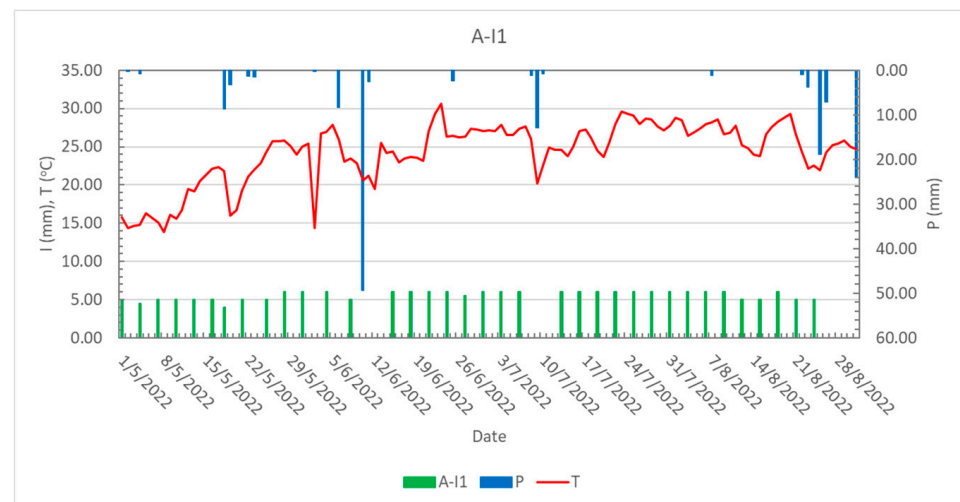
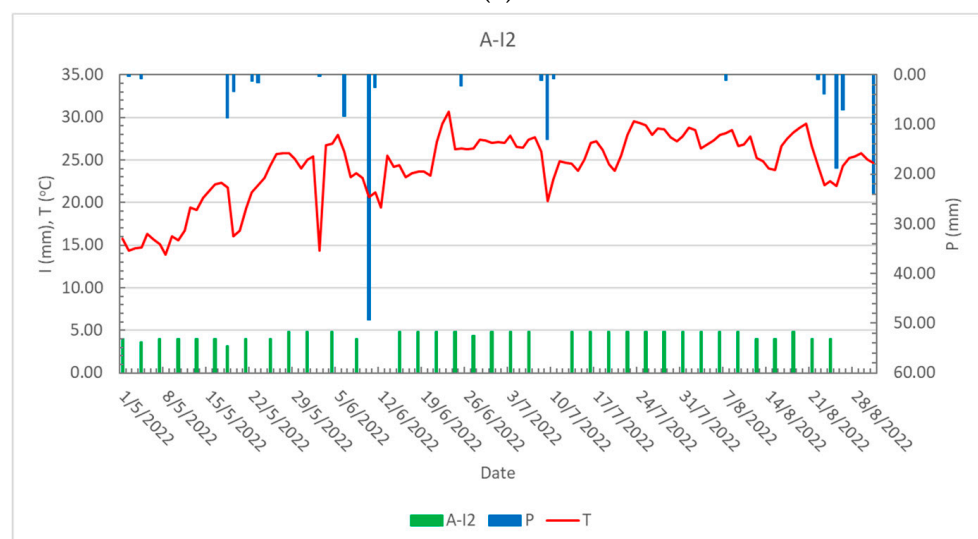


Figure 7. Temporal variation in soil moisture. (a) A-I1 treatment, (b) A-I2 treatment, (c) A-I3 treatment, (d) B-I1 treatment, (e) B-I2 treatment, and (f) B-I3 treatment.

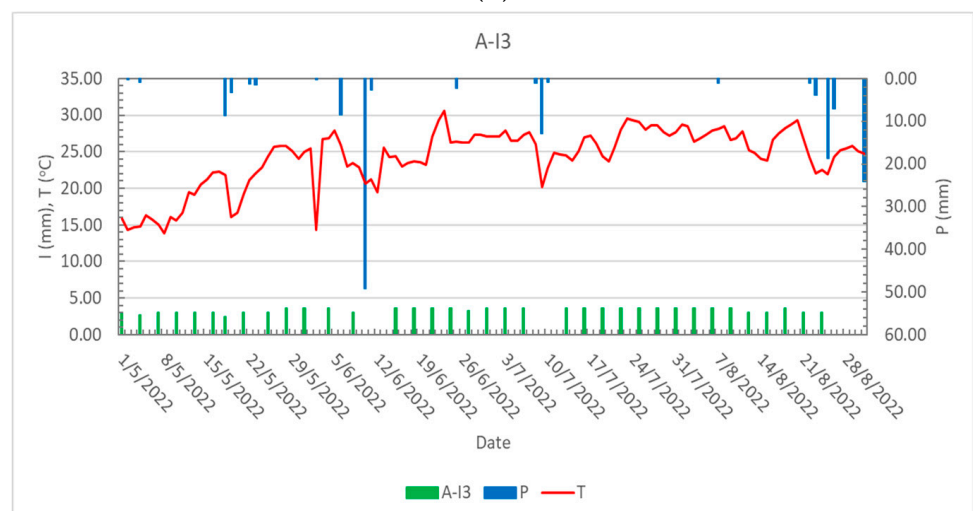
In Figures 9–11, the effect of the irrigation treatments on the quantitative characteristics (length, biomass and dry biomass) of the plant are presented. The figures present a comparative evaluation between each treatment and soil type, along with the standard deviation bars (a), the density of values and data distribution (b) and the statistical differences between the quantitative characteristics of the plant for each irrigation regime and soil type (c), correspondingly. Table 4 provides the details of the ANOVA statistical analysis.



(a)

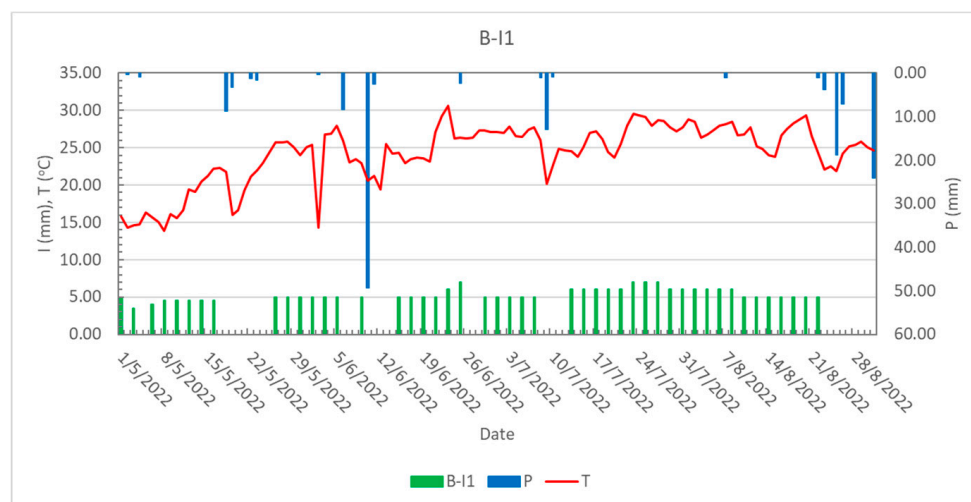


(b)

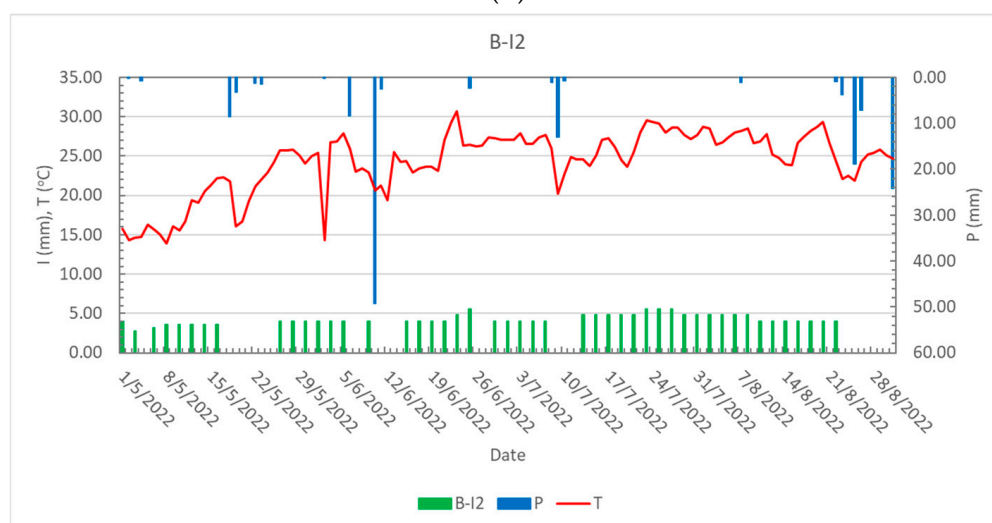


(c)

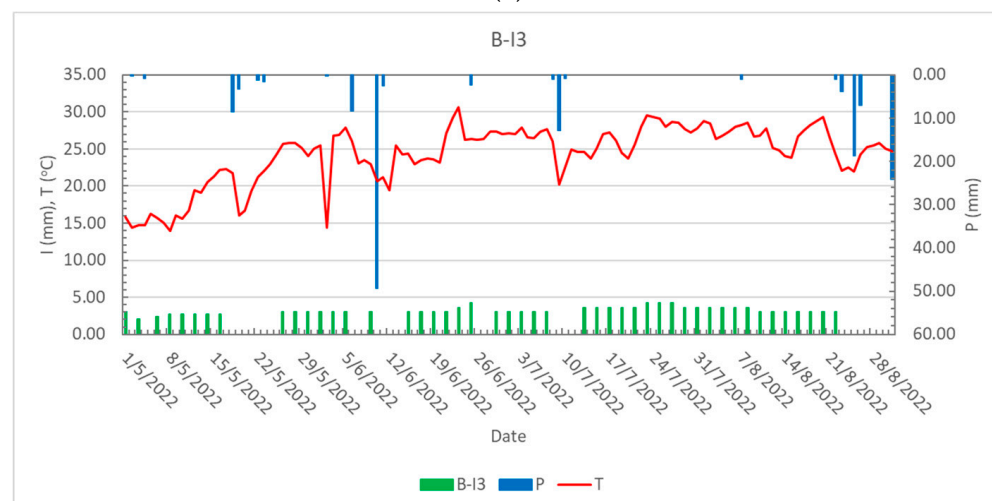
Figure 8. Cont.



(d)



(e)



(f)

Figure 8. Irrigation, precipitation and temperature. (a) A-I1 treatment, (b) A-I2 treatment, (c) A-I3 treatment, (d) B-I1 treatment, (e) B-I2 treatment, and (f) B-I3 treatment.

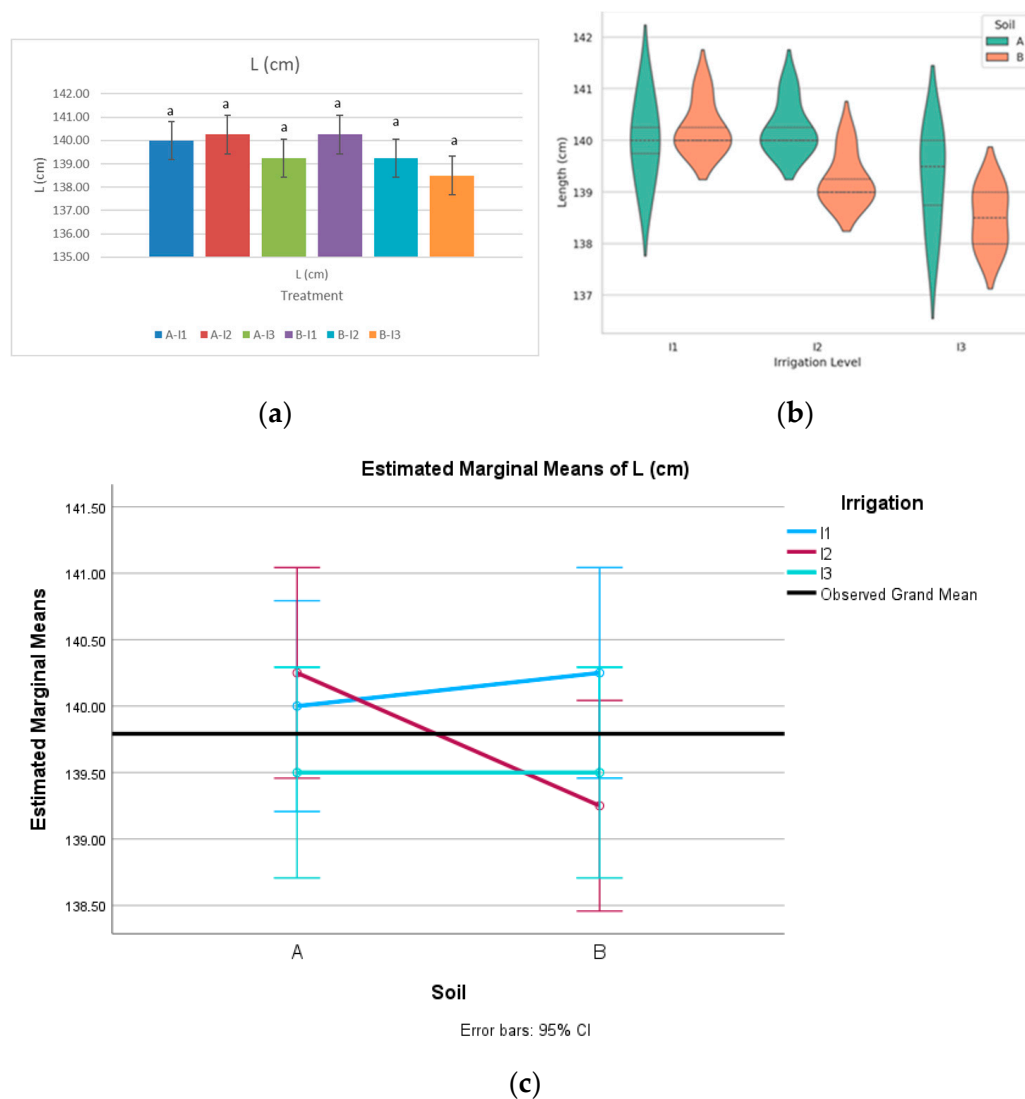


Figure 9. Plant length for all irrigation treatments for both soils (a), density of values and data distribution (b), and estimated marginal means of the plant length for the factors: irrigation and soil (c). Means followed by the same letter are not significantly different ($p > 0.05$).

Table 4. Analysis of variance (length, biomass, dry biomass).

Variable	Source of Variation	df	Sum of Squares	Mean Square	F-Value	p-Value
Length (L)	Irrigation (I)	2	1.583	0.792	1.390	0.275
	Soil (S)	1	0.375	0.375	0.659	0.428
	I $\times$ S	2	1.750	0.875	1.537	0.242
	Error	18	10.250	0.569		
Biomass (m)	Irrigation (I)	2	2.358	1.179	8.922	0.002
	Soil (S)	1	0.242	0.242	1.832	0.193
	I $\times$ S	2	0.004	0.002	0.017	0.983
	Error	18	2.378	0.132		
Dry Biomass (dry m)	Irrigation (I)	2	0.377	0.189	8.922	0.002
	Soil (S)	1	0.039	0.039	1.832	0.193
	I $\times$ S	2	0.001	0.000	0.017	0.983
	Error	18	0.381	0.021		

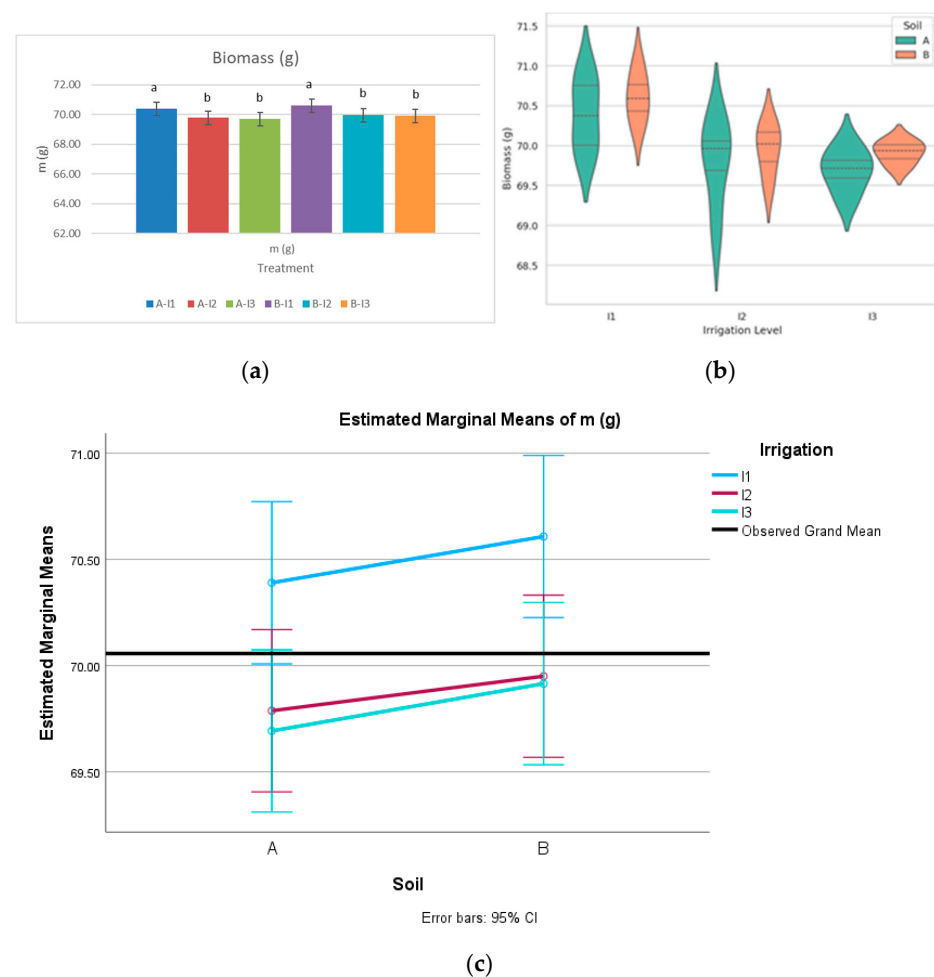


Figure 10. Plant biomass for all irrigation treatments for both soils (a), density of values and data distribution (b), estimated marginal means of the plant biomass for the factors: irrigation and soil (c). Different letters indicate statistically significant differences ($p < 0.05$).

To further demonstrate the effect of irrigation variance in fresh and dry biomass the eta squared ($\eta^2 = SS_{\text{irrigation}}/SS_{\text{total}}$) values are presented: 0.4733 for fresh biomass and 0.4724 for dry biomass indicating large effect ($\eta^2 > 0.14$), where $SS_{\text{irrigation}}$ is the sum of squares for irrigation and SS_{total} is the total sum of squares.

Figure 9 shows that the average values of length for each irrigation treatment and soil were as follows: A-I1: 140.00 cm, A-I2: 140.25 cm, A-I3: 139.25 cm, B-I1: 140.25 cm, B-I2: 139.25, and B-I3: 138.50 cm. There were no significant differences between all treatments ($p > 0.05$). Violin diagrams (Figure 9b) show the density of length values, along with the data distribution, where minor variation is shown, as most of the values are close to average. Figures 10 and 11 present the fresh and dry biomass across treatments and soil types. There are similar trends between fresh and dry biomass, as there were no significant differences between the I2 and I3 treatments ($p > 0.05$) (Figures 10a,c and 11a,c), whereas the control showed significantly higher biomass (fresh and dry). Data distribution showed minimal variance around the mean for both cases (Figures 10b and 11b).

In order to investigate the efficiency of water use, the irrigation water use efficiency (IWUE) was calculated as the ratio of the dry biomass to the total water applied for each irrigation level and the results are shown in Figure 12. Furthermore, in order to present the trend of the IWUE, taking into account the irrigation level along with the soil type, and because of the tendency of the IWUE to increase as the water input turns lower, the regression surface per soil and irrigation treatment was designed (Figure 13).

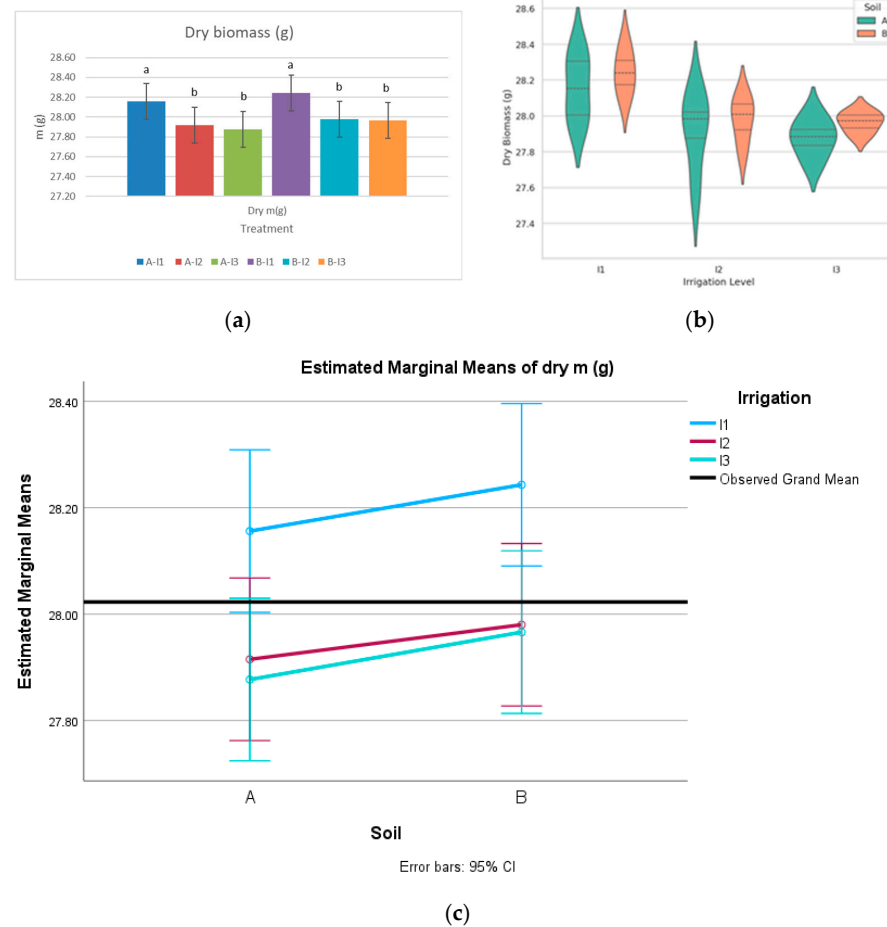


Figure 11. Plant dry biomass for all irrigation treatments for both soils (a), density of values and data distribution (b), and estimated marginal means of the plant dry biomass for the factors: irrigation and soil (c). Different letters indicate statistically significant differences ($p < 0.05$).

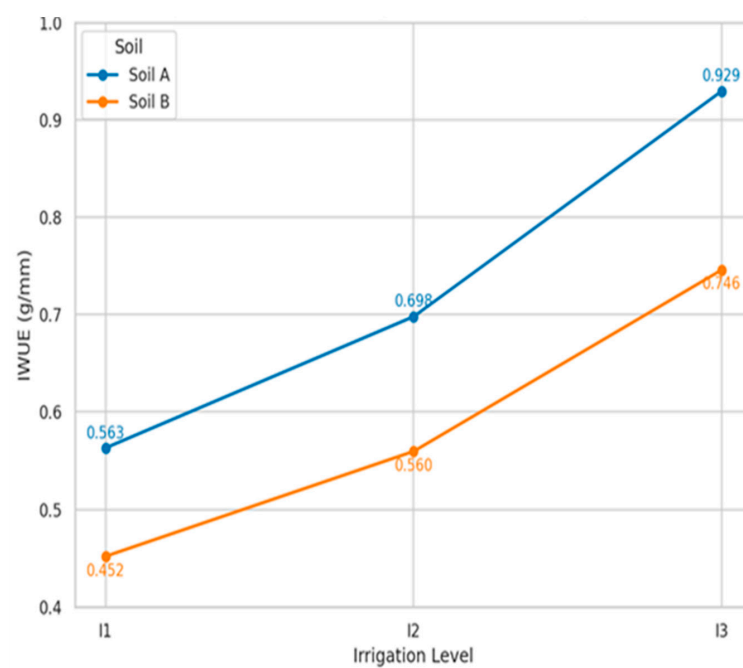


Figure 12. IWUE per soil and irrigation level.

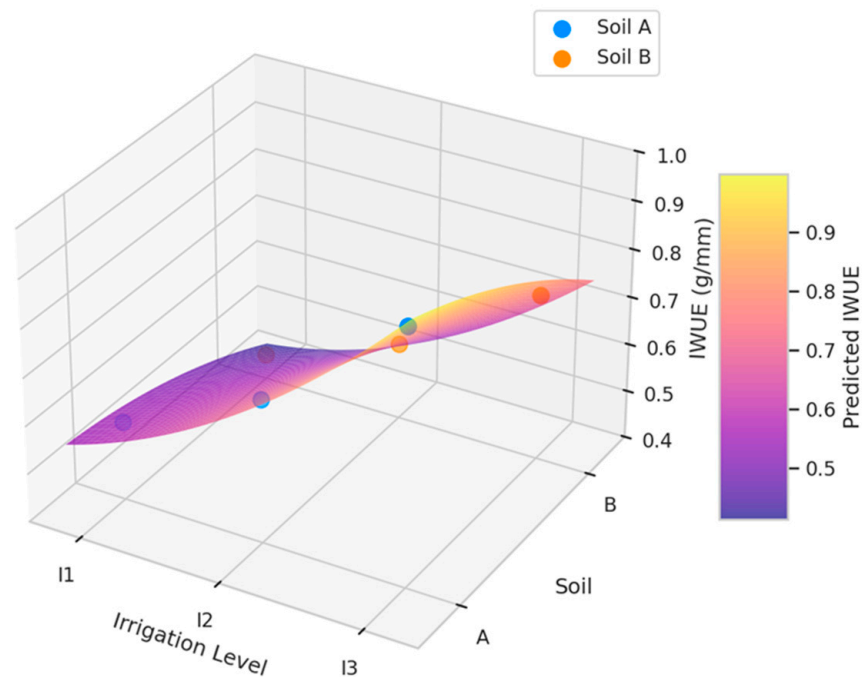


Figure 13. IWUE regression surface per soil and irrigation level.

The model equation of Figure 1, along with the R^2 value, is presented below:

$$z = 1.950580 - 0.0109453x - 0.0070760y + 0.0000199027x^2 + 0.0000454800(xy) - 0.0070760y^2 \quad (R^2 = 0.99823) \quad (7)$$

where

x = irrigation (mm).

y = soil type ($A = 0$, $B = 1$).

z = IWUE.

The model was developed to describe trends in the IWUE within the experimental irrigation and soil combinations. The variations in IWUE under specific irrigation and soil conditions are highlighted, while remaining within the framework set by the experimental procedure.

4. Discussion

Based on their respective classifications, clay loam (soil A) and sandy clay loam (soil B) showed distinct physical behaviors: soil A's finer texture allowed for greater water storage than soil B, which is a coarser texture soil. This behavior is reflected in the hydraulic properties of the two soils. Soil A showed greater soil moisture in saturation, residual moisture, FC and PWP, while bulk density and hydraulic conductivity were significantly lower. The difference between the soil textures was also captured at the cumulative infiltration curves and the infiltration rate curves (Figures 3 and 5), where at early times of infiltration, soil B showed a greater rate, as expected [18,32]. The infiltration rates of both soil samples were fundamental for the determination of the irrigation system used during the experiment. Soil A exhibited an initial infiltration rate of 2.5 mm/min (150 mm/h), reaching a steady state of 0.5 mm/min (30 mm/h). Soil B showed higher infiltrability, with initial and final values of 3.29 mm/min (197.4 mm/h) and 0.6 mm/min (36 mm/h), respectively. In order to avoid ponding, overflow and runoff, flow rate was set to 10 mm/h, considering the hydraulic behavior of soil A, providing a safety margin for both soil types. As infiltration is strongly influenced by both the soil texture and hydraulic parameters, it is

crucial to decode the water movement and define the hydraulic parameters of the soil at the soil–water system in order to achieve precise high-level irrigation management [33–36]. For this purpose, a deeper investigation into the hydraulic parameters has been achieved. The SWCC has been designed for both soils and was approximated with van Genuchten’s equation. Using the experimental suction–moisture pairs, more hydraulic parameters have been obtained, which was affected by the soil moisture, capturing the hydrodynamic behavior of each soil. In particular, the hydraulic capacity, hydraulic diffusivity, hydraulic conductivity, and pressure head behavior with the soil moisture was captured, showing that differences in soil texture significantly affect the irrigation-related hydraulic properties [37]. The above was considered to use water precisely in the irrigation system. Clay loam showed greater field capacity, while its SWCC showed greater availability of water for the plants (Figure 4). Thus, the selected MAD, equal to 0.6, guaranteed that irrigation was kept within the limits where soil moisture remained viable for the plant. The selected MAD ensured hydraulic security at soil B too (Figure 6), preventing sharp increases in suction that would bring enormous stress to the plant.

Hydraulic capacity $C(\theta)$ revealed the buffering capacity of CL soil, as the curve was significantly “smoother” than the one of SCL soil (Figures 4 and 6). The smaller slope of $C(\theta)$ indicates that soil A was less sensitive to moisture changes, signifying smoother water supply to the plant. Hydraulic capacity is defined as $d\theta/dh$, meaning that the sharp drop of $C(\theta)$ in soil B (Figure 6) indicates that as soon as the plant absorbs a small amount of water, the suction rises sharply, making the remaining water difficult to access [37,38]. More specifically, soil B shows a higher $C_{\max}$ of about 4 cm^{-1} compared to the $C_{\max}$ of 0.11 cm^{-1} of soil A, but drops drastically within a narrow frame of soil moisture. Thus, the hydraulic capacity derived from van Genuchten’s model served as a proxy for the hydraulic buffering ability of each soil. Moreover, $C(\theta)$ curves indicated that soil A revealed a more stable state for the water content than soil B, which justifies the decision for less frequent irrigation intervals at soil A, as it had the ability to supply the plants with the required amounts of water with fewer changes in suction, preventing transient water stress (Figures 7 and 8). The hydraulic capacity analysis indicated that in the soil moisture range, the peak value, along with the shape of the curve, is a key point for irrigation management. The distinct slopes and the narrow or wide peaks of the curves provide a mechanistic explanation for the soil’s retention ability and water availability to plants. Soil A acted as a reliable hydraulic buffer, supporting plants even at 60% FC (I3) treatment. On the contrary, soil B with the sharp curve of hydraulic capacity could provide water within a narrow range of soil moisture, demanding more intensive and careful water management. Such hydraulic behaviors can explain the observed differences in IWUE and the peak of A-I3 treatment. Also, the sharp hydraulic capacity of soil B demonstrated likely induced transient water stress, resulting in lower IWUE compared to soil A across all irrigation levels.

Figures 4 and 6 also provide the hydraulic diffusivity curves for both soils, indicating that as diffusivity is defined as $D(\theta) = K(\theta)/C(\theta)$, the impact of both hydraulic conductivity and capacity on the hydraulic diffusivity is significant [30]. As a result, soil A showed significantly higher diffusivity than soil B, indicating that the water profile was more uniform and steadier compared to soil B, where the lower values of $D(\theta)$ mean that the water movement was physically constrained, as a result of the soil’s texture, structure and hydraulic behavior [30,37]. Moreover, the significant difference in IWUE between the two soils is mathematically justified by the relationship between $D(\theta)$, $K(\theta)$ and $C(\theta)$, confirming the benefit of the soil physics–agricultural hydraulics perspective. Hence, soil A resulted in about 12-fold higher diffusivity ($D_{\max} \approx 0.23 \text{ cm}^2 \text{ min}^{-1}$) within a wider range of soil moisture compared to soil B ($D_{\max} \approx 0.02 \text{ cm}^2 \text{ min}^{-1}$), which facilitated the water redistribution of soil A within the pot, allowing the plant to maximize its water

uptake, even at the lowest water input (A-I3). Diffusivity analysis demonstrated that water redistribution of soil B was faster near saturation, but the selected supply (10 mm/h) was compatible with the need for mitigation of local saturation. The soil physics–agricultural hydraulics perspective of irrigation management, considering the analysis of the $I(t)$, $i(t)$, SWCC $h(\theta)$, $C(\theta)$, $K(\theta)$ and $D(\theta)$ curves, leads to precise irrigation scheduling and water application, as soil A allowed for more flexible scheduling due to its higher ability to retain and provide available water to plants. In contrast, soil B, due to its “flashy” hydraulic behavior, necessitated intensive and more careful irrigation management. The hydraulic performance of both soils was also captured at the $\theta(t)$ curves of Figure 7, where the quick draining of soil B is notable. Figure 8 shows the irrigation water applied, along with precipitation and temperature, where the differences in frequency and doses of water are indicated.

Regarding the plant length (L), Table 4 indicates that irrigation ($F_{2,18} = 1.390$, $p = 0.275$) and soil type ($F_{1,18} = 0.659$, $p = 0.428$) have no significant impact on plants. Furthermore, no significant interaction between irrigation and soil type was observed ($F_{2,18} = 1.537$, $p = 0.242$). Figure 9 shows that industrial hemp is a plant that is resistant to water stress [39–41], as the crop height was not significantly affected at both soil types. Figure 9a shows the comparative evaluation between each treatment and soil type, while in Figure 9b, the density of values, along with the data distribution, is presented with a minor variation, as most of the values are close to the average. The average values for the length of each irrigation treatment and soil were as follows: A-I1: 140.00 cm, A-I2: 140.25 cm, A-I3: 139.25 cm, B-I1: 140.25 cm, B-I2: 139.25, and B-I3: 138.50 cm. Figure 9c presents the statistical differences between the plant’s height (L) for each irrigation regime and soil type, for the significance level $\alpha = 0.05$. The length of the plants with treatments I2 and I3 did not show significant differences between them ($p > 0.05$), while there were no significant differences for both treatments compared to the control ($p > 0.05$). Industrial hemp is usable for fiber; hence, the height (as stems contain fiber) is very important as the source of fiber, while the total biomass refers to, e.g., leaves, flowers, etc., that are irrelevant for the textile quality. However, the approach of linking the plant’s height with the fiber yield provides an indirect assessment and this limitation should be mentioned. Regarding biomass (m and dry m), Table 4 reveals the significant effect of irrigation ($F_{2,18} = 8.922$, $p = 0.002$), while the soil type ($F_{1,18} = 1.832$, $p = 0.193$) and interaction of the two variables have no significant impact on plants ($F_{2,18} = 0.017$, $p = 0.983$). The same pattern was followed by dry biomass (irrigation: $F_{2,18} = 8.922$, $p = 0.002$, soil type: $F_{1,18} = 1.832$, $p = 0.193$, irrigation and soil type: $F_{2,18} = 0.017$, $p = 0.983$). Although the two soil types under investigation showed distinct hydraulic properties, having no significant differences on biomass (fresh and dry) between them suggests that the irrigation protocol followed standardized plant performance across different soils. In Figures 10 and 11, the fresh and dry biomass are presented by using the same visualization as for the length of the plants. Figures 10a and 11a show the comparative evaluation between each treatment and soil type regarding fresh and dry biomass, correspondingly. The violin diagrams (Figures 10b and 11b) represent the density of the values of fresh and dry biomass, correspondingly, along with the data distribution, where minor variation is shown, as most of the values are close to the average for both cases. At Figures 10c and 11c, fresh and dry biomass follow the same law, as there are no significant differences between treatments I2 and I3 ($p > 0.05$), while the control shows higher fresh and dry biomass values, as expected [7]. However, when biomass production is interpreted in relation to the irrigation water applied, the irrigation water use efficiency (IWUE) becomes a more informative indicator of the plant performance under water-limited conditions. The ability of the clay loam (soil A) to retain water led to higher irrigation use efficiency (IWUE), especially at the 60% FC level, where reducing the irrigation dose rather improved

the biomass to water consumption ratio. Figures 12 and 13 capture the tendencies of the IWUE for each irrigation level and soil type, showing that clay loam had the ability to efficiently provide water to the plant, even with the I3 (60% FC) treatment. Both figures also show that the IWUE increases as irrigation decreases for both soils, meaning that in both soil cases, less water led to higher water efficiency. However, clay loam soil gave a higher IWUE than sandy clay loam across the irrigation levels and the control (I1: 0.563 g/mm I2: 0.698 g/mm, I3: 0.929 g/mm). Moreover, the highest level of IWUE was observed when the irrigation level was low to moderate at the clay loam soil (surface peak of 0.929 g/mm at Figure 13), a finding that is linked to hydraulic capacity, as in the 60% FC range, soil A has a sufficient buffering capacity, using water more efficiently. For sandy clay loam (soil B), the efficiency was lower at all irrigation levels, so the trend shows that clay loam was more efficient overall, indicating that among the conditions of the containerized cultivation system, reduced irrigation can be applied to the point that the biomass does not drastically drop. The sandy clay loam soil was characterized by a higher hydraulic conductivity and infiltration rate, causing water to penetrate more rapidly while being retained less in the soil profile [18]. The hydraulic capacity $C(\theta)$ can explain the limited available soil moisture for plants and the fluctuations of the soil moisture. In addition, the diffusivity levels resulted in faster, but less stable, water movement in the rhizosphere, increasing the risk of unevenness and the need for more frequent irrigation events. Thus, irrigation management in the sandy clay loam soil required a more intensive schedule to avoid water stress beyond the required values, but also reduced the irrigation water use efficiency (I1: 0.452 g/mm, I2: 0.560 g/mm, I3: 0.746 g/mm) due to quick infiltration, indicating that sandy clay loam requires more careful water management. The performance of the soils in terms of the IWUE is directly linked to their hydraulic characteristics. The maximization of the IUWE in soil A under the I3 regime highlights the optimal utilization of water resources when considering hydraulic properties. The steeper curve of $C(\theta)$, along with the narrow soil moisture range, the lower diffusivity and the higher infiltrability of soil B, is responsible for the intense scheduling of water application and the lower efficiency of the irrigation water, meaning that hydraulic properties directly affect irrigation planning, water management and IWUE [11,42,43]. As a result, the clay loam soil offered greater flexibility, allowing for fewer, but more efficient irrigation applications, while the sandy clay loam required stricter and more frequent planning to meet either the crop needs or the reduced crop needs as planned. Despite the different strategies, the results showed that the three irrigation regimes of full and deficit irrigation did not significantly affect the productivity of Felina 32 in terms of height, yet the fresh and dry biomass were affected, as expected [7,9,39]. However, the increased irrigation water use efficiency in clay loam at 60% FC demonstrates that the rational combination of hydraulic properties with adapted irrigation strategies can achieve both production maintenance and water resource conservation [43–46], notably for applications in textile/fabric production.

The findings of the present study suggest that irrigation planning based on hydraulic properties reveals the potential of significant improvement of IWUE, highlighting the value of precision irrigation in environments with limited water resources, such as the Mediterranean. The findings are a proof-of-concept of how soil physics and agricultural hydraulics determine the sustainability of industrial hemp. The present study focuses on the research of inherent hydraulic properties in a controlled hydraulic environment, in order to monitor the water balance and simultaneously avoid deep percolation and runoff, and also minimize the spatial heterogeneity of the soil. It is important to mention the limitations of this research though, which refers to the certain variety of industrial hemp (Felina 32), the certain soil types, the containerized environment and the Mediterranean conditions of central Greece, while additional repetitions of the experiment in future work

would help to verify the observed trends. In any case, the hydraulic properties of the two soils under research played a decisive role in irrigation management and planning. As we move into times of increasing water scarcity due to climate change, especially in areas that face a water shortage even now, in the sensitive area of the Mediterranean [47], the precise irrigation water management practices will play an important role in the conservation of precious water resources.

5. Conclusions

The main scope of the present research was to integrate hydraulic properties into the irrigation practices of industrial hemp (Felina 32). For this purpose, two different soil types were used to investigate the impact of the different texture and hydraulic behavior of each soil on irrigation scheduling in full and deficit irrigation treatments, along with the effect on the irrigation water use efficiency. The main finding of the study is the mechanistic link between hydraulic properties and irrigation. Moreover, irrigation management based on hydraulic properties led to the following key findings:

- Irrigation system, optimal flow rate and MAD selection: infiltration analysis led to a select flow rate at a safety level below the final infiltrability in order to avoid ponding and runoff. SWCC and diffusivity provided the basis for the MAD selection and the spaghetti irrigation system, respectively.
- Hydraulic properties and irrigation decisions: Decoding the hydraulic behavior of each soil revealed their different ability for providing water to plants. The combined analysis of hydraulic capacity $C(\theta)$, hydraulic conductivity $K(\theta)$ and diffusivity $D(\theta)$ facilitated the separation of the irrigation strategy, as the soil moisture range of the curves and the peak value and the shape of the curves was a key point for irrigation management. As a result, the soil physics–agricultural hydraulics perspective of irrigation management allowed for more flexible scheduling for clay loam soil, due to its higher ability to retain and provide the available water to plants linked to the wider curve of $C(\theta)$, the higher $D(\theta)$ values, the lower $K(\theta)$ and the wider range of soil moisture. In contrast, sandy clay loam, due to its abrupt hydraulic behavior, necessitated an intensive and more careful irrigation management, as it showed a sharp $C(\theta)$ curve, lower $D(\theta)$ values and higher $K(\theta)$ in a narrower range of soil moisture.
- Impact on irrigation water use efficiency: Irrigation based on hydraulic properties highlighted the ability of clay loam to efficiently apply water to Felina 32 across full and deficit irrigation. The rise in IWUE with the decrease in water applied is directly linked to the soil's specific hydraulic properties, connecting soil physics and agricultural hydraulics with peak agronomic performance.

While the present research was conducted in a controlled containerized environment, the integration of hydraulic properties into irrigation decisions provides a mechanistic framework for future field-scale applications.

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